

IP Addressing and Network Segmentation Plan

A structured IP addressing scheme and network segmentation strategy are foundational components of a secure, scalable, and easily maintainable enterprise network. By grouping similar devices into dedicated subnets, an organization can effectively manage traffic flows, apply granular security policies, and optimize network performance.

1. IP Addressing and Subnet Allocation Table

The table below outlines the organization of the company network into distinct functional sections using a private Class C IPv4 addressing space (192.168.0.0/16). Each category is assigned a dedicated /24 subnet, allowing up to 254 usable host addresses per segment.

Device Category	Network Address (CIDR)	Usable Host Range	Subnet Mask	Description / Purpose
Servers	192.168.10.0/24	192.168.10.1 - 192.168.10.254	255.255.255.0	Critical infrastructure, active directory, databases, and local applications. Typically assigned static IPs.
User Devices	192.168.20.0/24	192.168.20.1 - 192.168.20.254	255.255.255.0	Wired corporate workstations, desktop PCs, and employee laptops. Dynamically assigned via DHCP.
Printers	192.168.30.0/24	192.168.30.1 - 192.168.30.254	255.255.255.0	Networked multi-function

Device Category	Network Address (CIDR)	Usable Host Range	Subnet Mask	Description / Purpose
				printers and copiers. Assigned static IPs or reserved DHCP addresses.
Wi-Fi	192.168.40.0/24	192.168.40.1 - 192.168.40.254	255.255.255.0	Wireless access points handling mobile devices, guest access, and wireless corporate assets.

2. Core Network Strategy and Rationale

Why Devices Are Separated

In a flat (unsegmented) network, all devices share a single broadcast domain. This means every computer, server, and printer receives every broadcast packet sent across the network, leading to unnecessary traffic congestion. Devices are separated into logical sections to reduce broadcast noise, control traffic patterns, and ensure that different device types do not interfere with one another. For example, high-volume print jobs or broadcast traffic from user workstations will not degrade the performance of critical server applications.

How Segmentation Improves Security

Network segmentation is an essential layer of defense-in-depth security. By placing boundaries between subnets, you prevent unrestricted lateral movement. If a single user device on the 192.168.20.0/24 network is compromised by malware or ransomware, the infection is isolated to that specific subnet. Firewalls and Access Control Lists (ACLs) can be placed between the subnets to restrict traffic. For instance, you can configure a policy that allows the *User Devices* subnet to access the *Printers* subnet only on specific printing ports, while completely blocking the untrusted *Wi-Fi* subnet from accessing internal database servers directly.

Easier Troubleshooting and Management

Organizing a network systematically simplifies day-to-day administrative tasks and rapid incident response:

- **Immediate Identification:** Network administrators can instantly identify a device's general type and location simply by looking at its IP address. An alert coming from 192.168.10.45 is immediately recognized as a server issue, whereas an alert from 192.168.40.112 points to a wireless client.
- **Isolating Fault Domains:** If a network loop occurs or a device misbehaves (e.g., a "broadcast storm"), the disruption is contained within that specific subnet, ensuring the rest of the company's business operations remain functional.
- **Tailored Technical Rules:** Administrators can easily apply customized configurations—such as Quality of Service (QoS) priorities for servers or unique DHCP lease durations for Wi-Fi users—without affecting other device categories.